

Git Hands on Learning -3

Branching:

Creating a Branch & Switch Branch:

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git branch GitNewBranch

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ ls
application.log  log/  welcome.txt

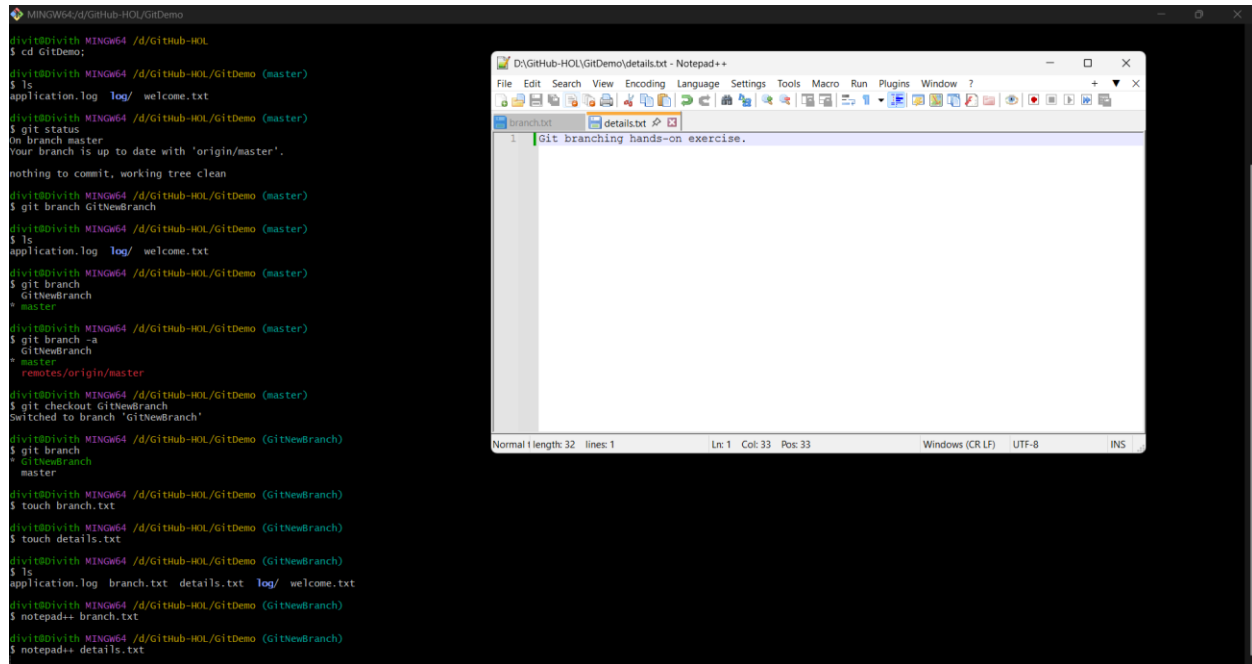
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git branch
  GitNewBranch
* master

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git branch -a
  GitNewBranch
* master
remotes/origin/master

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git checkout GitNewBranch
Switched to branch 'GitNewBranch'

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (GitNewBranch)
$ git branch
* GitNewBranch
  master
```

Adding + Modifying the Files



The image shows a terminal window on the left and a Notepad++ editor on the right. The terminal window displays the following commands and output:

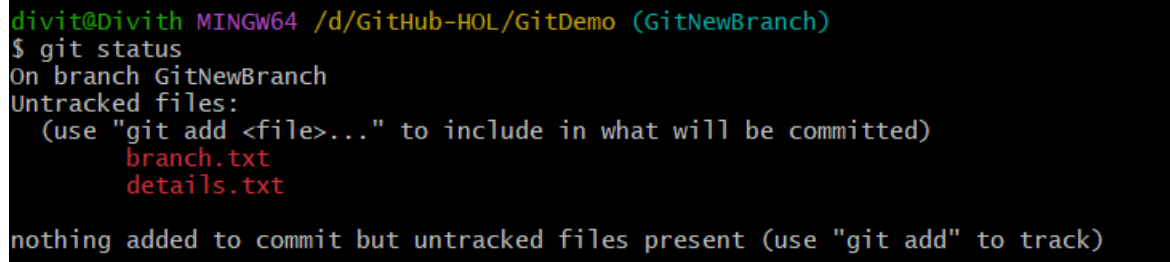
```
divit@Divith MINGW64 /d/GitHub-HOL
$ cd GitDemo;
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ ls
application.log  log/  welcome.txt
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git branch GitNewBranch
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ ls
application.log  log/  welcome.txt
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git branch
* GitNewBranch
  master
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git branch -a
GitNewBranch
* master
  remotes/origin/master
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git checkout GitNewBranch
Switched to branch 'GitNewBranch'
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (GitNewBranch)
$ git branch
* GitNewBranch
  master
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (GitNewBranch)
$ touch branch.txt
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (GitNewBranch)
$ touch details.txt
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (GitNewBranch)
$ ls
application.log  branch.txt  details.txt  log/  welcome.txt
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (GitNewBranch)
$ notepad++ branch.txt
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (GitNewBranch)
$ notepad++ details.txt
```

The Notepad++ editor on the right shows the file `details.txt` with the following content:

```
1 | Git branching hands-on exercise.
```

Git Status Information:

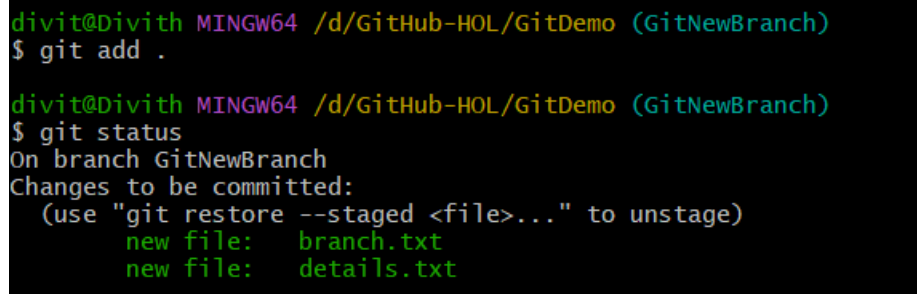


The image shows a terminal window with the following output:

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (GitNewBranch)
$ git status
On branch GitNewBranch
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    branch.txt
    details.txt

nothing added to commit but untracked files present (use "git add" to track)
```

Adding in the Git:



The image shows a terminal window with the following output:

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (GitNewBranch)
$ git add .
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (GitNewBranch)
$ git status
On branch GitNewBranch
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   branch.txt
    new file:   details.txt
```

Git Commit Status:

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (GitNewBranch)
$ git commit -m "Added files in GitNewBranch"
[GitNewBranch 39ac866] Added files in GitNewBranch
 2 files changed, 2 insertions(+)
 create mode 100644 branch.txt
 create mode 100644 details.txt

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (GitNewBranch)
$ git status
On branch GitNewBranch
nothing to commit, working tree clean
```

Merging:

Switching to Master Branch:

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (GitNewBranch)
$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git branch
  GitNewBranch
* master
```

Difference of Master & GitNewBranch

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git diff master GitNewBranch
diff --git a/branch.txt b/branch.txt
new file mode 100644
index 0000000..8f3257e
--- /dev/null
+++ b/branch.txt
@@ -0,0 +1 @@
+This file was created in GitNewBranch.
\ No newline at end of file
diff --git a/details.txt b/details.txt
new file mode 100644
index 0000000..842b449
--- /dev/null
+++ b/details.txt
@@ -0,0 +1 @@
+Git branching hands-on exercise.
\ No newline at end of file
```

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git diff --stat master GitNewBranch
branch.txt | 1 +
details.txt | 1 +
2 files changed, 2 insertions(+)
```

Merging the Both GitNewBranch & Master.

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git merge GitNewBranch
Updating f6a6bcd..39ac866
Fast-forward
branch.txt | 1 +
details.txt | 1 +
2 files changed, 2 insertions(+)
create mode 100644 branch.txt
create mode 100644 details.txt
```

Files in the Branch

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ ls
application.log  branch.txt  details.txt  log/  welcome.txt
```

Logs:

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git log --oneline --graph --decorate
* 39ac866 (HEAD -> master, GitNewBranch) Added files in GitNewBranch
* f6a6bcd (origin/master) Added gitignore to ignore log files
* a7fa51c Added welcome.txt file
```

Deleting the Branch

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was 39ac866).

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git branch
* master
```